

1. Questions

Study the following information carefully and the following question.

The given table DI below shows the number of players who participated in the Commonwealth Games, the percentage of the number of players who won medals out of the total number of players who participated, and the ratio of the sum of the number of males and females who participated to the difference between the number of males and females who participated in the Commonwealth Games in four different countries i.e. I, P, N and A.

Country	The total number of players who participated	The percentage of the number of players who won medals out of the total number of players who participated	(Male + Female): (Male – Female)
I	$X + 20$	77.5%	15: 1
P	$Y - 10$	35%	7: 1
N	$Z - 15$	40%	37: 7
A	-	$(0.4Y)\%$	11: 3

Notes:

- i). The total number of players (male + female) who participated = The number of players who won medals + the number of players who didn't win medals
- ii). The number of players who didn't win medals in countries I and P is 27 and 91, respectively.
- iii). The number of female participants in countries P and A are equal.
- iv). In each country, the number of males who participated is more than that of females.

If the values of X, Y, and Z are arithmetic progression (X is a smaller value and Z is a larger value). Find the number of female participants in country N is what percentage of that in country P. Which of the following statement (s) is/are true?

I: $(X + 25)\%$

II: $(Y - 25)\%$

III: $(Z - 75)\%$

- a. Only I
- b. Both I and II
- c. Both II and III
- d. Both I and III
- e. All I, II, and III

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2. Questions

If 57.14% of the male participants won the medals from country A, find the ratio of the number of female players who didn't win medals from country A to the number of players who won medals from country P.

- a. 3:7
- b. 4:7
- c. 1:3
- d. 5:9
- e. 3:8

3. Questions

If the number of players who won medals in country M is $Y/2$ and the number of players who didn't win medals in country M is $X/4$. If the number of male participants from country M is 45% of the total number of players who participated in country M. Find the average of the number of female participants in countries I, A, and M.

- a. 53
- b. 63
- c. 57
- d. 65
- e. 43

4. Questions

The difference between the number of players who didn't win medals from countries I and P is what percentage of the difference between the number of male participants from countries A and P?

- a. $(2X + 56)\%$
- b. $(2Y - 44)\%$
- c. $(X + 98)\%$
- d. Both (a) and (b)
- e. Both (b) and (c)

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5. Questions

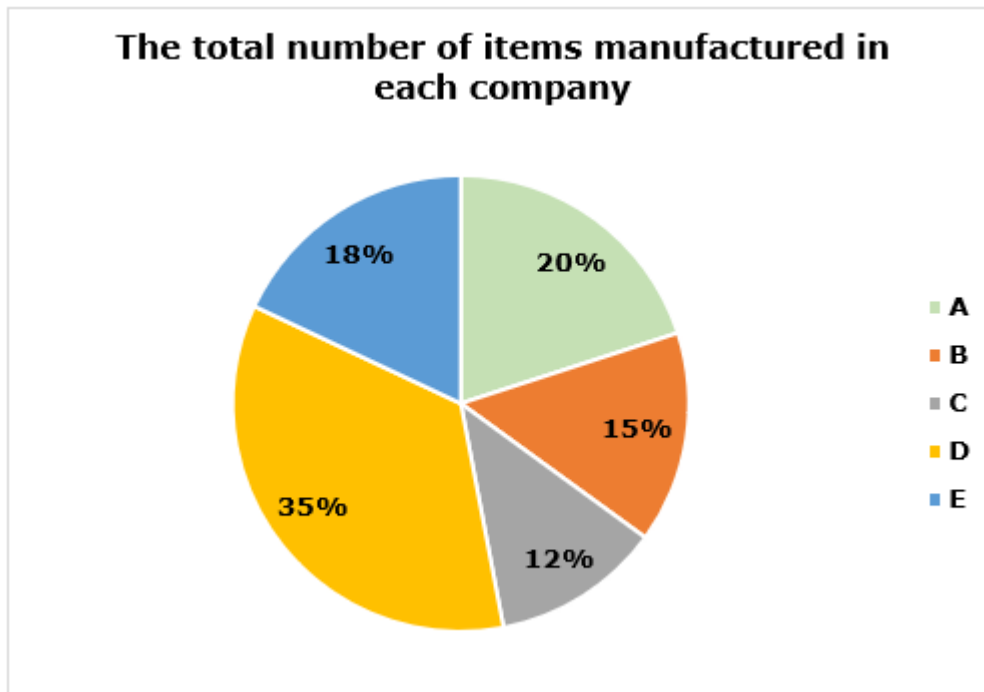
If the difference between the number of players who won medals in country N and the number of players who didn't win medals from country N is 37. If the number of male participants who won the medals from country N is equal to that of female participants, find the number of female players who didn't win medals from country N.

- a. 27
- b. 38
- c. 37
- d. 48
- e. None of these

6. Questions

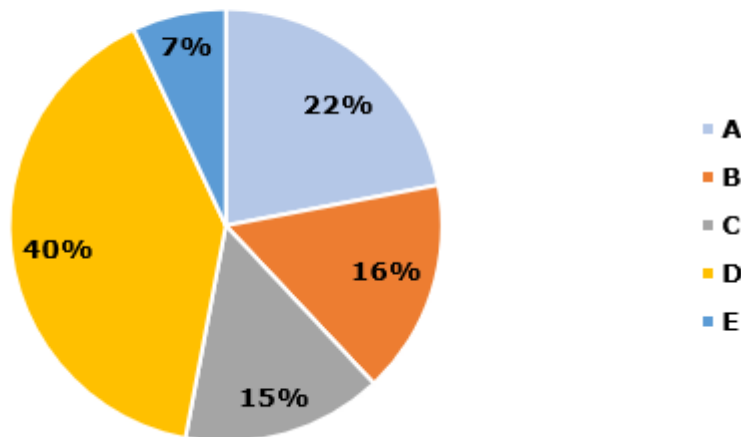
Study the following information carefully and the following question.

In 2022, the given first pie chart below shows the percentage distribution of the total number of items manufactured in each company out of the total number of items manufactured in five companies (i.e. A, B, C, D and E) together. The given second pie chart shows below the percentage distribution of the total number of items sold in each company out of the total number of items sold in five companies together.



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The total number of items sold in each company



Notes:

i). The total number of items unsold in five companies together is 44.44% of the total number of items manufactured in five companies together.

ii). The total number of items unsold in five companies together is a perfect cube is T^3 .

iii). $2T + R = 80$ and $4T + 5R = 280$.

The average of the number of items manufactured by companies B, E, and F is 2500. If the number of items sold by company F is $33\frac{1}{3}\%$ that of company C, then find the number of unsold items by company F.

- a. 960
- b. 1260
- c. 1060
- d. 560
- e. 760

7. Questions

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The total number of items manufactured by Company A in 2023 increased by 27.5% compared to the previous year. If the total number of unsold items by Company A in 2023 remains the same as in 2022, find the percentage increase in the number of items sold by Company A in 2023 compared to the previous year.

- a. 45%
- b. 40%
- c. 37.5%

d. 42.5%

e. 32.5%

8. Questions

In Company D, 37.5% of the items sold are plastic, which is 33.33% more than the number of plastic items unsold and remaining are non-plastic items. Find the ratio of the number of non-plastic items manufactured in Company D and the number of items manufactured in Company A.

a. 45:41

b. 33:41

c. 29:48

d. 49:40

e. 49:48

9. Questions

If the cost price of an item in company E is Rs. 400 and the total selling amount of all sold items in company E is Rs. 294000. Find the profit percentage of an item in company E. (Only sold items are considered for profit.)

a. 10%

b. 15%

c. 7.5%

d. 5%

e. 2.5%

10. Questions

The total number of items manufactured by company X is 40% less than that by company B, and the ratio of the number of items unsold by company X to that by company D is 7:23, respectively. If the number of items sold by company X is P, find the value of 'P + 80'.

I: It's a perfect cube

II: It's a perfect square

III: It's multiple of 10

a. Both I and II

b. Both I and III

c. Both II and III

d. None is true

e. None of these

11. Questions

Study the following information carefully and the following question.

The average of the present ages of A, B, and C is 'X' years. The ratio of A's age 5 years ago to B's age 2 years hence is 5:8. The present ages of A, B, and C are 'a', 'b', and 'c' years, respectively. The present age of C is 66.66% of the present age of B.

All three invested some amount in Schemes P and Q:

Scheme P: It offers 'c'% simple interest per annum. The interest received by A after 3 years is Rs. 60X. The amounts invested by A, B, and C are Rs. 2500, Rs. 3200, and Rs. 4800, respectively.

Scheme Q: It offers 'b/3'% compound interest annually. The amounts invested by A, B, and C are Rs. 100b, Rs. 100X, and Rs. 8cX, respectively.

Note: The investment time period in scheme P is the same for all persons and the investment time period in scheme Q is the same for all persons.

If the interest received by A from scheme Q is Rs. 993, find the total amount received by C in schemes P and Q together after the respective time period.

- a. Rs. 11264
- b. Rs. 8004
- c. Rs. 13004
- d. Rs. 3204
- e. Cannot be determined

12. Questions

If the present age of D is 'c'% less than that of B. The average of the present ages of A, C, and D is 'M' years. Which of the following statement(s) is/are true?

I: $X < M$

II: $X > M$

III: $X = M$

- a. Only II
- b. Only I
- c. Only III
- d. Both I and III
- e. Both II and III

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13. Questions

If the interest received by C from both scheme P and scheme Q together is Rs. 3720. Find the invested time period for C in scheme Q.

- a. 3 years
- b. 4 years
- c. 1 year
- d. 2 years
- e. Cannot be determined

14. Questions

If the average ages of A, B, C, D and E after 5 years will be $(X + 3)$ years and the age of D, before 3 years was 40% less than the age of E after 3 years. The present age of D is what percentage of the sum of the age of A after 3 years and the present age of E?

- a. 22%
- b. 30%
- c. 32.5%
- d. 36%
- e. 44.44%

15. Questions

If the ratio of the interest received by B from schemes P and Q is 128:35, respectively. Find the ratio of the interest received by B from scheme Q to that by A.

- a. 4:5
- b. 2:3
- c. 3:4
- d. 7:8
- e. 5:6

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16. Questions

The cost price and selling price of an article are Rs. a and Rs. $2a$, respectively, and the article is marked up by Rs. 900 above the cost price. If both the selling price and cost price of the article are increased by Rs. 200, while the marked price remains the same, find the difference between the new discount and the new profit on the article, given that $a = 28^2 - 384$.

- a. Rs. 150
- b. Rs. 100
- c. Rs. 250

d. Rs. 200

e. Rs. 300

17. Questions

A, B, and C together can complete a work in $M/9$ days, while A and B together can complete the same work in $M/6$ days. If B alone can complete the work in $M/2$ days, and C takes 5 days more than A to complete the work, then find the time taken by B to complete $M/2.5$ of the work.

a. 24 days

b. 16 days

c. 8 days

d. 20 days

e. 12 days

18. Questions

David invested Rs. X and Mano invested Rs. 9000 in a business. After five months, David increased his initial investment by 50%, and after two more months, Mano decreased his investment by 30%. At the end of a year, the profit share of David is Rs. 2480 out of a total profit of Rs. 5000. Which of the following statement (s) is/are true?

I: The value of $\sqrt{(X - 71)}$ is a multiple of 11.

II: The initial investment of David is 33.33% of Mano's investment.

III: X is an even number.

a. Both I and III

b. Only I

c. Both II and III

d. Only II

e. Both I and II

19. Questions

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If 399 is subtracted from a three-digit number, the resultant is 85. The unit digit of the three-digit number is ___ less than its tens place digit. If the three-digit number is reversed, the difference between the original and the reversed number is ____.

Find the missing values.

a. 4 and 2

b. 2 and 0

c. 4 and 4

d. 4 and 0

e. 8 and 4

20. Questions

The time taken by train A, whose length is 'd' meters, to cross a 300-meter-long platform is equal to the time taken by train B, whose length is 'd+200' meters, to cross a 500-meter-long platform. If the ratio of the speed of train A to that of train B is 7:11, then which of the following statement(s) is/are true according to the find the value of 'd' (in meters).

- a. It is a perfect square
- b. It is a multiple of 4
- c. It is a three-digit number
- d. Both (a) and (b)
- e. All (a), (b), and (c)

Explanations:

1. Questions

Country I,

The number of players who didn't win medals = 27

The number of players who won medals = $27 * (77.5/22.5) = 93$

The total number of players who participated = $27 + 93 = 120$

$$X + 20 = 120$$

$$X = 100$$

The sum of the number of male and female participants = 120

The difference between the number of male and female participants = $120 * (1/5) = 8$

So,

The number of male participants = $(120 + 8)/2 = 128/2 = 64$

The number of female participants = $120 - 64 = 56$

Country P,

The number of players who didn't win medals = 91

The number of players who won medals = $91 * (35/65) = 49$

The total number of players who participated = $91 + 49 = 140$

$$Y - 10 = 140$$

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$$Y = 150$$

The sum of the number of male and female participants = 140

The difference between the number of male and female participants = $140 * (1/7) = 20$

So,

The number of male participants = $(140 + 20)/2 = 160/2 = 80$

The number of female participants = $140 - 80 = 60$

Country A,

According to the note iii),

The number of female participants = 60

$(\text{Male} + \text{Female})/(\text{Male} - \text{Female}) = 11/3$

$(\text{Male} + 60)/(\text{Male} - 60) = 11/3$

$3 * \text{Male} + 180 = 11 * \text{Male} - 660$

$8 * \text{Male} = 840$

Male = 105

The number of male participants = 105

The total number of players who participated = $105 + 60 = 165$

The number of players who won medals = $(0.4Y)\% 165 = 60\% \text{ of } 165 = 99$

The number of players who didn't win medals = $165 - 99 = 66$

Countries	The total number of players who participated	The number of players who won medals	The number of players who didn't win medals	The number of male participants	The number of female participants
I	120	93	27	64	56
P	140	49	91	80	60
N	-	-	-	-	-
A	165	99	66	105	60

Answer: E

Let according to the question,

Formula: arithmetic progression, $a_n = a + (n - 1) * d$

$a_n = Z$

$a = X = 100$

$n = 3$

$$d = 150 - 100 = 50$$

Then,

$$Z = 100 + (3 - 1) * 50$$

$$Z = 200$$

In country N,

$$\text{The total number of players who participated} = 200 - 15 = 185$$

$$\text{The difference between the number of male and female participants} = 185 * (7/37) = 35$$

$$\text{The number of male participants} = (185 + 35)/2 = 220/2 = 110$$

$$\text{The number of female participants} = 185 - 110 = 75$$

$$\text{Thus, percentage} = (75/60) * 100 = 125\%$$

$$\text{I: } (X + 25)\% = (100 + 25)\% = 125\% \text{ (Satisfied)}$$

$$\text{II: } (Y - 25)\% = (150 - 25)\% = 125\% \text{ (Satisfied)}$$

$$\text{III: } (Z - 75)\% = (200 - 75)\% = 125\% \text{ (Satisfied)}$$

Hence, option E

2. Questions

Country I,

$$\text{The number of players who didn't win medals} = 27$$

$$\text{The number of players who won medals} = 27 * (77.5/22.5) = 93$$

$$\text{The total number of players who participated} = 27 + 93 = 120$$

$$X + 20 = 120$$

$$X = 100$$

$$\text{The sum of the number of male and female participants} = 120$$

$$\text{The difference between the number of male and female participants} = 120 * (1/5) = 8$$

So,

$$\text{The number of male participants} = (120 + 8)/2 = 128/2 = 64$$

$$\text{The number of female participants} = 120 - 64 = 56$$

Country P,

$$\text{The number of players who didn't win medals} = 91$$

$$\text{The number of players who won medals} = 91 * (35/65) = 49$$

$$\text{The total number of players who participated} = 91 + 49 = 140$$

$$Y - 10 = 140$$

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$$Y = 150$$

The sum of the number of male and female participants = 140

The difference between the number of male and female participants = $140 * (1/7) = 20$

So,

The number of male participants = $(140 + 20)/2 = 160/2 = 80$

The number of female participants = $140 - 80 = 60$

Country A,

According to the note iii),

The number of female participants = 60

$(\text{Male} + \text{Female})/(\text{Male} - \text{Female}) = 11/3$

$(\text{Male} + 60)/(\text{Male} - 60) = 11/3$

$3 * \text{Male} + 180 = 11 * \text{Male} - 660$

$8 * \text{Male} = 840$

Male = 105

The number of male participants = 105

The total number of players who participated = $105 + 60 = 165$

The number of players who won medals = $(0.4Y)\% \text{ } 165 = 60\% \text{ of } 165 = 99$

The number of players who didn't win medals = $165 - 99 = 66$

Countries	The total number of players who participated	The number of players who won medals	The number of players who didn't win medals	The number of male participants	The number of female participants
I	120	93	27	64	56
P	140	49	91	80	60
N	-	-	-	-	-
A	165	99	66	105	60

Answer: A

The number of male participants who won the medals from country A = $57.14\% \text{ of } 105 = (4/7) * 105 = 60$

The number of female participants who won the medals from country A = $99 - 60 = 39$

The number of female participants who didn't win medals from country A = $60 - 39 = 21$

Thus, ratio = $21 : 49 = 3 : 7$

3. Questions

Country I,

The number of players who didn't win medals = 27

The number of players who won medals = $27 * (77.5/22.5) = 93$

The total number of players who participated = $27 + 93 = 120$

$$X + 20 = 120$$

$$X = 100$$

The sum of the number of male and female participants = 120

The difference between the number of male and female participants = $120 * (1/5) = 8$

So,

The number of male participants = $(120 + 8)/2 = 128/2 = 64$

The number of female participants = $120 - 64 = 56$

Country P,

The number of players who didn't win medals = 91

The number of players who won medals = $91 * (35/65) = 49$

The total number of players who participated = $91 + 49 = 140$

$$Y - 10 = 140$$

$$Y = 150$$

The sum of the number of male and female participants = 140

The difference between the number of male and female participants = $140 * (1/7) = 20$

So,

The number of male participants = $(140 + 20)/2 = 160/2 = 80$

The number of female participants = $140 - 80 = 60$

Country A,

According to the note iii),

The number of female participants = 60

$$(\text{Male} + \text{Female})/(\text{Male} - \text{Female}) = 11/3$$

$$(\text{Male} + 60)/(\text{Male} - 60) = 11/3$$

$$3 * \text{Male} + 180 = 11 * \text{Male} - 660$$

$$8 * \text{Male} = 840$$

$$\text{Male} = 105$$

The number of male participants = 105

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The total number of players who participated = $105 + 60 = 165$

The number of players who won medals = $(0.4Y)\% 165 = 60\%$ of $165 = 99$

The number of players who didn't win medals = $165 - 99 = 66$

Countries	The total number of players who participated	The number of players who won medals	The number of players who didn't win medals	The number of male participants	The number of female participants
I	120	93	27	64	56
P	140	49	91	80	60
N	-	-	-	-	-
A	165	99	66	105	60

Answer: C

The number of players who won medals in country M = $Y/2 = 150/2 = 75$

The number of players who didn't win medals in country M = $X/4 = 100/4 = 25$

The total number of players who participated in country M = $75 + 25 = 100$

The number of female participants from country M = $\{1 - (45/100)\} * 100 = 55$

Thus, average = $(56 + 60 + 55)/3 = 171/3 = 57$.

4. Questions

Country I,

The number of players who didn't win medals = 27

The number of players who won medals = $27 * (77.5/22.5) = 93$

The total number of players who participated = $27 + 93 = 120$

$X + 20 = 120$

$X = 100$

The sum of the number of male and female participants = 120 @QuizByMickey

The difference between the number of male and female participants = $120 * (1/5) = 8$

So,

The number of male participants = $(120 + 8)/2 = 128/2 = 64$

The number of female participants = $120 - 64 = 56$

Country P,

The number of players who didn't win medals = 91

The number of players who won medals = $91 * (35/65) = 49$

The total number of players who participated = $91 + 49 = 140$

$$Y - 10 = 140$$

$$Y = 150$$

The sum of the number of male and female participants = 140

The difference between the number of male and female participants = $140 * (1/7) = 20$

So,

The number of male participants = $(140 + 20)/2 = 160/2 = 80$

The number of female participants = $140 - 80 = 60$

Country A,

According to the note iii),

The number of female participants = 60

$$(\text{Male} + \text{Female})/(\text{Male} - \text{Female}) = 11/3$$

$$(\text{Male} + 60)/(\text{Male} - 60) = 11/3$$

$$3 * \text{Male} + 180 = 11 * \text{Male} - 660$$

$$8 * \text{Male} = 840$$

$$\text{Male} = 105$$

The number of male participants = 105

The total number of players who participated = $105 + 60 = 165$

The number of players who won medals = $(0.4Y)\% \text{ } 165 = 60\% \text{ of } 165 = 99$

The number of players who didn't win medals = $165 - 99 = 66$

Countries	The total number of players who participated	The number of players who won medals	The number of players who didn't win medals	The number of male participants	The number of female participants
I	120	93	27	64	56
P	140	49	91	80	60
N	-	-	-	-	-
A	165	99	66	105	60

Answer: D

The difference between the number of players who didn't win medals from countries I and P = $91 - 27 = 64$

The difference between the number of male participants from countries A and P = $105 - 80 = 25$

Thus, percentage = $(64/25) * 100 = 256\%$

a). $(2X + 56)\% = (200 + 56)\% = 256\%$ (this is satisfied)

b). $(2Y - 44)\% = (300 - 44)\% = 256\%$ (this is satisfied)

c). $(X + 98)\% = (100 + 98)\% = 198\%$ (this is not satisfied)

Therefore, Both (a) and (b)

Hence, option D

5. Questions

Country I,

The number of players who didn't win medals = 27

The number of players who won medals = $27 * (77.5/22.5) = 93$

The total number of players who participated = $27 + 93 = 120$

$$X + 20 = 120$$

$$X = 100$$

The sum of the number of male and female participants = 120

The difference between the number of male and female participants = $120 * (1/5) = 8$

So,

The number of male participants = $(120 + 8)/2 = 128/2 = 64$

The number of female participants = $120 - 64 = 56$

Country P,

The number of players who didn't win medals = 91

The number of players who won medals = $91 * (35/65) = 49$

The total number of players who participated = $91 + 49 = 140$

$$Y - 10 = 140$$

$$Y = 150$$

The sum of the number of male and female participants = 140

The difference between the number of male and female participants = $140 * (1/7) = 20$

So,

The number of male participants = $(140 + 20)/2 = 160/2 = 80$

The number of female participants = $140 - 80 = 60$

Country A,

According to the note iii),

@QuizByMickey

The number of female participants = 60

$(\text{Male} + \text{Female})/(\text{Male} - \text{Female}) = 11/3$

$(\text{Male} + 60)/(\text{Male} - 60) = 11/3$

$3 * \text{Male} + 180 = 11 * \text{Male} - 660$

$8 * \text{Male} = 840$

Male = 105

The number of male participants = 105

The total number of players who participated = $105 + 60 = 165$

The number of players who won medals = $(0.4Y)\% 165 = 60\%$ of $165 = 99$

The number of players who didn't win medals = $165 - 99 = 66$

Countries	The total number of players who participated	The number of players who won medals	The number of players who didn't win medals	The number of male participants	The number of female participants
I	120	93	27	64	56
P	140	49	91	80	60
N	-	-	-	-	-
A	165	99	66	105	60

Answer: B

The percentage of the number of players who won medals in country N = 40%

The percentage of the number of players who didn't win medals from country N = $100\% - 40\% = 60\%$

$60\% - 40\% = 20\%$

$20\% = 37$

The number of players who won medals in country N = $37 * (40/20) = 74$

The number of players who didn't win medals from country N = $37 * (60/20) = 111$

Let the number of male participants who won the medals from country N be m

So, the number of female participants who won the medals from country N = m

$m + m = 74$

$2m = 74$

$m = 37$

The number of female participants who won the medals from country N = 37

The total number of players who participated from country N = $74 + 111 = 185$

The difference between the number of male and female participants from country N = $185 * (7/37) = 35$

The number of male participants from country N = $(185 + 35)/2 = 220/2 = 110$

The number of female participants from country N = $185 - 110 = 75$

The number of female players who didn't win medals from country N = $75 - 37 = 38$

6. Questions

WKT, the

From note iii),

Solving the equations, we get

$R = 40$ and $T = 20$.

So, the value = $T^3 = 8000$.

The total number of items unsold in five companies together = 8000

The total number of items manufactured in five companies together = $8000 * (100/44.44) = 8000 * (9/4) = 18000$

The total number of items sold in five companies together = $18000 - 8000 = 10000$

In company A,

The number of items manufactured = 20% of 18000 = 3600

The number of items sold = 22% of 10000 = 2200

The number of items unsold = $3600 - 2200 = 1400$

In company B,

The number of items manufactured = 15% of 18000 = 2700

The number of items = 16% of 10000 = 1600

The number of items unsold = $2700 - 1600 = 1100$

In company C,

The number of items manufactured = 12% of 18000 = 2160

The number of items sold = 15% of 10000 = 1500

The number of items unsold = $2160 - 1500 = 660$

In company D,

The number of items manufactured = 35% of 18000 = 6300

The number of items sold = 40% of 10000 = 4000

The number of items unsold = $6300 - 4000 = 2300$

In company E,

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The number of items manufactured = 18% of 18000 = 3240

The number of items sold = 7% of 10000 = 700

The number of items unsold = 3240 – 700 = 2540

Companies	The total number of items manufactured	The number of items sold	The number of unsold items
A	3600	2200	1400
B	2700	1600	1100
C	2160	1500	660
D	6300	4000	2300
E	3240	700	2540

Answer: C

The sum of the number of items manufactured by companies B, E, and F together = 3 * 2500 = 7500

The total number of items manufactured by company F = 7500 – 2700 – 3240 = 1560

The number of items sold by company F = 33(1/3)% of 1500 = 1500 * (1/3) = 500

The number of items unsold by company F = 1560 – 500 = 1060

7. Questions

WKT, the

From note iii),

Solving the equations, we get

R = 40 and T = 20.

So, the value = $T^3 = 8000$.

The total number of items unsold in five companies together = 8000

The total number of items manufactured in five companies together = 8000 * (100/44.44) = 8000 * (9/4) = 18000

The total number of items sold in five companies together = 18000 – 8000 = 10000

In company A,

The number of items manufactured = 20% of 18000 = 3600

The number of items sold = 22% of 10000 = 2200

The number of items unsold = 3600 – 2200 = 1400

In company B,

The number of items manufactured = 15% of 18000 = 2700

The number of items = 16% of 10000 = 1600

The number of items unsold = 2700 – 1600 = 1100

In company C,

The number of items manufactured = 12% of 18000 = 2160

The number of items sold = 15% of 10000 = 1500

The number of items unsold = 2160 – 1500 = 660

In company D,

The number of items manufactured = 35% of 18000 = 6300

The number of items sold = 40% of 10000 = 4000

The number of items unsold = 6300 – 4000 = 2300

In company E,

The number of items manufactured = 18% of 18000 = 3240

The number of items sold = 7% of 10000 = 700

The number of items unsold = 3240 – 700 = 2540

Companies	The total number of items manufactured	The number of items sold	The number of unsold items
A	3600	2200	1400
B	2700	1600	1100
C	2160	1500	660
D	6300	4000	2300
E	3240	700	2540

Answer: A

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The total number of items manufactured by company A in 2023 = $(127.5/100) \times 3600 = 4590$

The number of items unsold by company A in 2023 = 1400

The number of items sold by company A in 2023 = 4590 – 1400 = 3190

Thus, percentage = $[(3190 - 2200)/2200] \times 100 = 990/22 = 45\%$

8. Questions

WKT, the

From note iii),

Solving the equations, we get

$R = 40$ and $T = 20$.

So, the value = $T^3 = 8000$.

The total number of items unsold in five companies together = 8000

The total number of items manufactured in five companies together = $8000 * (100/44.44) = 8000 * (9/4) = 18000$

The total number of items sold in five companies together = $18000 - 8000 = 10000$

In company A,

The number of items manufactured = 20% of 18000 = 3600

The number of items sold = 22% of 10000 = 2200

The number of items unsold = $3600 - 2200 = 1400$

In company B,

The number of items manufactured = 15% of 18000 = 2700

The number of items = 16% of 10000 = 1600

The number of items unsold = $2700 - 1600 = 1100$

In company C,

The number of items manufactured = 12% of 18000 = 2160

The number of items sold = 15% of 10000 = 1500

The number of items unsold = $2160 - 1500 = 660$

In company D,

The number of items manufactured = 35% of 18000 = 6300

The number of items sold = 40% of 10000 = 4000

The number of items unsold = $6300 - 4000 = 2300$

In company E,

The number of items manufactured = 18% of 18000 = 3240

The number of items sold = 7% of 10000 = 700

The number of items unsold = $3240 - 700 = 2540$

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Companies	The total number of items manufactured	The number of items sold	The number of unsold items
A	3600	2200	1400
B	2700	1600	1100
C	2160	1500	660
D	6300	4000	2300
E	3240	700	2540

Answer: E

The number of plastic items sold in company D = $4000 * (37.5/100) = 1500$

The number of plastic items unsold in company D = $1500 * (3/4) = 1125$

The number of non – plastic items sold in company D = $4000 - 1500 = 2500$

The number of non – plastic items unsold in company D = $2300 - 1125 = 1175$

The number of non – plastic items manufactured in company D = $2500 + 1175 = 3675$

The required ratio = $3675: 3600 = 49:48$

9. Questions

WKT, the

From note iii),

Solving the equations, we get

$R = 40$ and $T = 20$.

So, the value = $T^3 = 8000$.

The total number of items unsold in five companies together= 8000

The total number of items manufactured in five companies together = $8000 * (100/44.44) = 8000 * (9/4) = 18000$

The total number of items sold in five companies together = $18000 - 8000 = 10000$

In company A,

The number of items manufactured= 20% of $18000 = 3600$

The number of items sold = 22% of $10000 = 2200$

The number of items unsold = $3600 - 2200 = 1400$

In company B,

The number of items manufactured= 15% of $18000 = 2700$

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The number of items = 16% of 10000 = 1600

The number of items unsold = 2700 – 1600 = 1100

In company C,

The number of items manufactured= 12% of 18000 = 2160

The number of items sold = 15% of 10000 = 1500

The number of items unsold = 2160 – 1500 = 660

In company D,

The number of items manufactured= 35% of 18000 = 6300

The number of items sold = 40% of 10000 = 4000

The number of items unsold = 6300 – 4000 = 2300

In company E,

The number of items manufactured= 18% of 18000 = 3240

The number of items sold = 7% of 10000 = 700

The number of items unsold = 3240 – 700 = 2540

Companies	The total number of items manufactured	The number of items sold	The number of unsold items
A	3600	2200	1400
B	2700	1600	1100
C	2160	1500	660
D	6300	4000	2300
E	3240	700	2540

Answer: D

The cost price of an item in company E = Rs. 400

The selling price of an item in company E = 294000/700 = Rs. 420

Thus, profit percentage = $[(420 - 400)/400] * 100 = 5\%$

10. Questions

WKT, the

From note iii),

Solving the equations, we get

R = 40 and T = 20.

@QuizByMickey

So, the value = $T^3 = 8000$.

The total number of items unsold in five companies together = 8000

The total number of items manufactured in five companies together = $8000 * (100/44.44) = 8000 * (9/4) = 18000$

The total number of items sold in five companies together = $18000 - 8000 = 10000$

In company A,

The number of items manufactured = 20% of 18000 = 3600

The number of items sold = 22% of 10000 = 2200

The number of items unsold = $3600 - 2200 = 1400$

In company B,

The number of items manufactured = 15% of 18000 = 2700

The number of items = 16% of 10000 = 1600

The number of items unsold = $2700 - 1600 = 1100$

In company C,

The number of items manufactured = 12% of 18000 = 2160

The number of items sold = 15% of 10000 = 1500

The number of items unsold = $2160 - 1500 = 660$

In company D,

The number of items manufactured = 35% of 18000 = 6300

The number of items sold = 40% of 10000 = 4000

The number of items unsold = $6300 - 4000 = 2300$

In company E,

The number of items manufactured = 18% of 18000 = 3240

The number of items sold = 7% of 10000 = 700

The number of items unsold = $3240 - 700 = 2540$

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Companies	The total number of items manufactured	The number of items sold	The number of unsold items
A	3600	2200	1400
B	2700	1600	1100
C	2160	1500	660
D	6300	4000	2300
E	3240	700	2540

Answer: B

The total number of items manufactured by company X = $2700 * \{1 - (40/100)\} = (60/100) * 2700 = 1620$

The number of items unsold in company X = $2300 * (7/23) = 700$

The number of items sold in company X = $1620 - 700 = 920$

$P = 920$

$P + 80 = 920 + 80 = 1000$ (It's a perfect cube and It's multiple of 10)

Therefore, both I and III is true.

Hence, option B.

11. Questions

WKT, the present ages of A, B, and C are 'a', 'b', and 'c' years respectively.

$$a + b + c = 3X \rightarrow (1)$$

Let the age of A, 5 years ago be 5y years.

So, the age of B, 2 years hence = $5y * (8/5) = 8y$ years

The present age of A = $a = (5y + 5)$ years

The present age of B = $b = (8y - 2)$ years

The present age of C = $c = (66.66/100) * (8y - 2) = [2(8y - 2)/3] = [(16y - 4)/3]$ years

From (1),

$$5y + 5 + 8y - 2 + [(16y - 4)/3] = 3X$$

$$15y + 15 + 24y - 6 + 16y - 4 = 9X$$

$$55y + 5 = 9X \rightarrow (2)$$

In scheme P, the interest received by A = 60X

Let,

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$$2500 * c/100 * 3 = 60X$$

$$75c = 60X$$

$$c/X = 4/5$$

WKT,

$$c = [(16y - 4)/3] \text{ years}$$

$$[(16y - 4)/3]/X = 4/5$$

$$(4y - 1)/X = 3/5$$

$$X = (5/3) * (4y - 1)$$

$$X = (20y - 5)/3 \text{ à (3)}$$

The value of X, apply on equation (1), we get

$$55y + 5 = 9 * [(20y - 5)/3]$$

$$55y + 5 = 60y - 15$$

$$5y = 20$$

$$y = 4$$

and,

$$X = ((20 * 4) - 5)/3$$

$$X = 75/3$$

$$X = 25$$

Let,

The present age of A = a = (5y + 5) = 20 + 5 = 25 years

$$a = 25$$

The present age of B = b = (8y - 2) = 32 - 2 = 30 years

$$b = 30$$

The present age of C = c = [(16y - 4)/3] = 60/3 = 20 years

$$c = 20$$

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The interest received by B in scheme P = $3200 * 3 * (20/100)$ = Rs. 1920

The interest received by C in scheme P = $4800 * 3 * (20/100)$ = Rs. 2880

In Scheme Q,

It offered 'b/3'% compounded annually. So, the interest rate is (30/3 = 10) 10%

The amount invested by A = $100 * 30$ = Rs. 3000

The amount invested by B = $100 * 25$ = Rs. 2500

The amount invested by C = $8 * 20 * 25 = \text{Rs. } 4000$

Answer: C

According to the question,

$$CI = P \{ [1 + (R/100)]^n - 1 \}$$

Let the time period for A in scheme Q be n years

$$3000 * [110/100]^n - 3000 = 993$$

$$3000 * [11/10]^n = 3993$$

$$[11/10]^n = 1331/1000$$

$$[11/10]^n = [11/10]^3$$

$$n = 3$$

Let,

$$\text{The amount received by C from scheme Q} = 4000 * [110/100]^3$$

$$= 4000 * [1331/1000]$$

$$= 4 * 1331 = \text{Rs. } 5324$$

$$\text{The amount received by C in scheme P} = 4800 + 2880 = \text{Rs. } 7680$$

$$\text{Thus, required sum} = 5324 + 7680 = \text{Rs. } 13004$$

12. Questions

WKT, the present ages of A, B, and C are 'a', 'b', and 'c' years respectively.

$$a + b + c = 3X \rightarrow (1)$$

Let the age of A, 5 years ago be 5y years.

$$\text{So, the age of B, 2 years hence} = 5y * (8/5) = 8y \text{ years}$$

$$\text{The present age of A} = a = (5y + 5) \text{ years}$$

$$\text{The present age of B} = b = (8y - 2) \text{ years}$$

$$\text{The present age of C} = c = (66.66/100) * (8y - 2) = [2(8y - 2)/3] = [(16y - 4)/3] \text{ years}$$

From (1),

$$5y + 5 + 8y - 2 + [(16y - 4)/3] = 3X$$

$$15y + 15 + 24y - 6 + 16y - 4 = 9X$$

$$55y + 5 = 9X \rightarrow (2)$$

$$\text{In scheme P, the interest received by A} = 60X$$

Let,

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$$2500 * c/100 * 3 = 60X$$

$$75c = 60X$$

$$c/X = 4/5$$

WKT,

$$c = [(16y - 4)/3] \text{ years}$$

$$[(16y - 4)/3]/X = 4/5$$

$$(4y - 1)/X = 3/5$$

$$X = (5/3) * (4y - 1)$$

$$X = (20y - 5)/3 \text{ à (3)}$$

The value of X, apply on equation (1), we get

$$55y + 5 = 9 * [(20y - 5)/3]$$

$$55y + 5 = 60y - 15$$

$$5y = 20$$

$$y = 4$$

and,

$$X = ((20 * 4) - 5)/3$$

$$X = 75/3$$

$$X = 25$$

Let,

The present age of A = a = (5y + 5) = 20 + 5 = 25 years

$$a = 25$$

The present age of B = b = (8y - 2) = 32 - 2 = 30 years

$$b = 30$$

The present age of C = c = [(16y - 4)/3] = 60/3 = 20 years

$$c = 20$$

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The interest received by B in scheme P = $3200 * 3 * (20/100)$ = Rs. 1920

The interest received by C in scheme P = $4800 * 3 * (20/100)$ = Rs. 2880

In Scheme Q,

It offered 'b/3'% compounded annually. So, the interest rate is (30/3 = 10) 10%

The amount invested by A = $100 * 30$ = Rs. 3000

The amount invested by B = $100 * 25$ = Rs. 2500

The amount invested by C = $8 * 20 * 25 = \text{Rs. } 4000$

Answer: A

The present age of D = $(100 - c)\%$ of 30 = 80% of 30 = 24 years

The average of the present ages of A, C, and D = $(25 + 20 + 24)/3 = 69/3 = 23$ years

M = 23

Here, X = 25

X > M is true.

Therefore, Only II

Hence, option A

13. Questions

WKT, the present ages of A, B, and C are 'a', 'b', and 'c' years respectively.

$$a + b + c = 3X \rightarrow (1)$$

Let the age of A, 5 years ago be 5y years.

So, the age of B, 2 years hence = $5y * (8/5) = 8y$ years

The present age of A = $a = (5y + 5)$ years

The present age of B = $b = (8y - 2)$ years

The present age of C = $c = (66.66/100) * (8y - 2) = [2(8y - 2)/3] = [(16y - 4)/3]$ years

From (1),

$$5y + 5 + 8y - 2 + [(16y - 4)/3] = 3X$$

$$15y + 15 + 24y - 6 + 16y - 4 = 9X$$

$$55y + 5 = 9X \rightarrow (2)$$

In scheme P, the interest received by A = 60X

Let,

$$2500 * c/100 * 3 = 60X$$

$$75c = 60X$$

$$c/X = 4/5$$

WKT,

$$c = [(16y - 4)/3] \text{ years}$$

$$[(16y - 4)/3]/X = 4/5$$

$$(4y - 1)/X = 3/5$$

$$X = (5/3) * (4y - 1)$$

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$$X = (20y - 5)/3 \text{ à } (3)$$

The value of X, apply on equation (1), we get

$$55y + 5 = 9 * [(20y - 5)/3]$$

$$55y + 5 = 60y - 15$$

$$5y = 20$$

$$y = 4$$

and,

$$X = ((20 * 4) - 5)/3$$

$$X = 75/3$$

$$X = 25$$

Let,

The present age of A = a = (5y + 5) = 20 + 5 = 25 years

$$a = 25$$

The present age of B = b = (8y - 2) = 32 - 2 = 30 years

$$b = 30$$

The present age of C = c = [(16y - 4)/3] = 60/3 = 20 years

$$c = 20$$

The interest received by B in scheme P = $3200 * 3 * (20/100)$ = Rs. 1920

The interest received by C in scheme P = $4800 * 3 * (20/100)$ = Rs. 2880

In Scheme Q,

It offered 'b/3'% compounded annually. So, the interest rate is (30/3 = 10) 10%

The amount invested by A = $100 * 30$ = Rs. 3000

The amount invested by B = $100 * 25$ = Rs. 2500

The amount invested by C = $8 * 20 * 25$ = Rs. 4000

Answer: D

The interest received by C from scheme Q = $3720 - 2880$ = Rs. 840

Let the invested time period for C from scheme Q be n years

$$4000 * [110/100]^n - 4000 = 840$$

$$[110/100]^n = 4840/4000$$

$$[11/10]^n = 121/100$$

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$$[11/10]^n = [11/10]^2$$

$$n = 2$$

The invested time period for C in scheme Q = 2 years

14. Questions

WKT, the present ages of A, B, and C are 'a', 'b', and 'c' years respectively.

$$a + b + c = 3X \rightarrow (1)$$

Let the age of A, 5 years ago be 5y years.

So, the age of B, 2 years hence = $5y * (8/5) = 8y$ years

The present age of A = a = (5y + 5) years

The present age of B = b = (8y - 2) years

The present age of C = c = $(66.66/100) * (8y - 2) = [2(8y - 2)/3] = [(16y - 4)/3]$ years

From (1),

$$5y + 5 + 8y - 2 + [(16y - 4)/3] = 3X$$

$$15y + 15 + 24y - 6 + 16y - 4 = 9X$$

$$55y + 5 = 9X \rightarrow (2)$$

In scheme P, the interest received by A = 60X

Let,

$$2500 * c/100 * 3 = 60X$$

$$75c = 60X$$

$$c/X = 4/5$$

WKT,

$$c = [(16y - 4)/3] \text{ years}$$

$$[(16y - 4)/3]/X = 4/5$$

$$(4y - 1)/X = 3/5$$

$$X = (5/3) * (4y - 1)$$

$$X = (20y - 5)/3 \rightarrow (3)$$

The value of X, apply on equation (1), we get

$$55y + 5 = 9 * [(20y - 5)/3]$$

$$55y + 5 = 60y - 15$$

$$5y = 20$$

$$y = 4$$

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and,

$$X = ((20 * 4) - 5)/3$$

$$X = 75/3$$

$$X = 25$$

Let,

The present age of A = a = (5y + 5) = 20 + 5 = 25 years

$$a = 25$$

The present age of B = b = (8y - 2) = 32 - 2 = 30 years

$$b = 30$$

The present age of C = c = [(16y - 4)/3] = 60/3 = 20 years

$$c = 20$$

The interest received by B in scheme P = $3200 * 3 * (20/100)$ = Rs. 1920

The interest received by C in scheme P = $4800 * 3 * (20/100)$ = Rs. 2880

In Scheme Q,

It offered 'b/3'% compounded annually. So, the interest rate is (30/3 = 10) 10%

The amount invested by A = $100 * 30$ = Rs. 3000

The amount invested by B = $100 * 25$ = Rs. 2500

The amount invested by C = $8 * 20 * 25$ = Rs. 4000

Answer: D

The average ages of A, B, C, D and E after 5 years = (25 + 3) = 28 years

The sum of the present ages of A, B, C, D and E after 5 years = (28 * 5) - (5 * 5) = 140 - 25 = 115 years

The sum of the present ages of D and E = 115 - 25 - 30 - 20 = 40 years

The ratio of the age of D, before 3 years to the age of E after 3 years = 60:100 = 3:5

Let the age of D, before 3 years and the age of E after 3 years be 3x years and 5x years respectively.

According to the question,

$$3x + 3 + 5x - 3 = 40$$

$$8x = 40$$

$$x = 5$$

The present age of D = $3x + 3 = 15 + 3 = 18$ years

The present age of E = $5x - 3 = 25 - 3 = 22$ years

The age of A after 3 years = $25 + 3 = 28$ years

The required percentage = $\{18 / (22 + 28)\} * 100 = 36\%$

15. Questions

WKT, the present ages of A, B, and C are 'a', 'b', and 'c' years respectively.

$$a + b + c = 3X \rightarrow (1)$$

Let the age of A, 5 years ago be 5y years.

So, the age of B, 2 years hence = $5y * (8/5) = 8y$ years

The present age of A = $a = (5y + 5)$ years

The present age of B = $b = (8y - 2)$ years

The present age of C = $c = (66.66/100) * (8y - 2) = [2(8y - 2)/3] = [(16y - 4)/3]$ years

From (1),

$$5y + 5 + 8y - 2 + [(16y - 4)/3] = 3X$$

$$15y + 15 + 24y - 6 + 16y - 4 = 9X$$

$$55y + 5 = 9X \rightarrow (2)$$

In scheme P, the interest received by A = 60X

Let,

$$2500 * c/100 * 3 = 60X$$

$$75c = 60X$$

$$c/X = 4/5$$

WKT,

$$c = [(16y - 4)/3] \text{ years}$$

$$[(16y - 4)/3]/X = 4/5$$

$$(4y - 1)/X = 3/5$$

$$X = (5/3) * (4y - 1)$$

$$X = (20y - 5)/3 \rightarrow (3)$$

The value of X, apply on equation (1), we get

$$55y + 5 = 9 * [(20y - 5)/3]$$

$$55y + 5 = 60y - 15$$

$$5y = 20$$

$$y = 4$$

and,

$$X = ((20 * 4) - 5)/3$$

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$$X = 75/3$$

$$X = 25$$

Let,

The present age of A = a = (5y + 5) = 20 + 5 = 25 years

$$a = 25$$

The present age of B = b = (8y - 2) = 32 - 2 = 30 years

$$b = 30$$

The present age of C = c = [(16y - 4)/3] = 60/3 = 20 years

$$c = 20$$

The interest received by B in scheme P = $3200 * 3 * (20/100)$ = Rs. 1920

The interest received by C in scheme P = $4800 * 3 * (20/100)$ = Rs. 2880

In Scheme Q,

It offered 'b/3'% compounded annually. So, the interest rate is (30/3 = 10) 10%

The amount invested by A = $100 * 30$ = Rs. 3000

The amount invested by B = $100 * 25$ = Rs. 2500

The amount invested by C = $8 * 20 * 25$ = Rs. 4000

Answer: E

The interest received by B from scheme Q = $1920 * (35/128)$ = 525

According to the question,

$$2500 * (110/100)^n - 2500 = 525$$

$$2500 * (11/10)^n = 3025$$

$$(11/10)^n = 121/100$$

$$(11/10)^n = (11/10)^2$$

$$n = 2$$

The interest received by A from scheme Q = $3000 * (110/100)^2 - 3000$

$$= 3000 * (21/100) = \text{Rs. } 630$$

Thus, ratio = 525:630 = 5:6

16. Questions

Answer: B

$$a = 28^2 - 384 = 784 - 384$$

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$$a = 400$$

The cost price of the article = Rs. 400

The selling price of the article = $2 * 400 = \text{Rs. } 800$

The marked price of the article = $400 + 900 = \text{Rs. } 1300$

Let,

The new cost price of the article = $400 + 200 = \text{Rs. } 600$

The new SP of the article = $800 + 200 = \text{Rs. } 1000$

Then,

The new discount of the article = $1300 - 1000 = \text{Rs. } 300$

The new profit of the article = $1000 - 600 = \text{Rs. } 400$

Thus, difference = $400 - 300 = \text{Rs. } 100$

17. Questions

Answer: E

Let assume the total work be M units

The efficiency of (A + B + C) = $M/(M/9) = 9 \text{ units/ day}$

The efficiency of (A + B) = $M/(M/6) = 6 \text{ units/ day}$

The efficiency of B = $M/(M/2) = 2 \text{ units/ day}$

The efficiency of A = $6 - 2 = 4 \text{ units/ day}$

The efficiency of C = $9 - 6 = 3 \text{ units/ day}$

The efficiency ratio of A: B: C = 4: 2: 3

The time ratio of A: B: C = $(1/4): (1/2): (1/3) = (3/12): (6/12): (4/12) = 3: 6: 4$

Let the time taken by A, B and C be 3x, 6x and 4x respectively.

The time taken by C to complete the work is 5 days more than that of A

$$4x - 3x = 5$$

$$x = 5$$

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The time taken by A, B, and C alone can complete the work is 15 days, 30 days and 20 days respectively.

M = LCM of 15, 30 and 20 = 60 units

The time taken by B alone can complete M/2.5 of the work = $(60/2.5)/2 = 24/2 = 12 \text{ days}$

18. Questions

Answer: A

The profit ratio of David and Mano = $[(X * 5) + (1.5X * 7)]: [(9000 * 7) + (70\% \text{ of } 9000 * 5)]$

$$= [5X + 10.5X]: [63000 + 31500] = 15.5X: 94500 = 3.1X: 18900$$

According to the question,

$$3.1X/(3.1X + 18900) = 2480/5000$$

$$X/(3.1X + 18900) = 800/5000$$

$$25X = 12.4X + 75600$$

$$12.6X = 75600$$

$$X = 6000$$

I: The value of $\sqrt{(X - 71)}$ is a multiple of 11.

$$\text{The required value} = \sqrt{(6000 - 71)} = \sqrt{5929} = 77$$

This statement is satisfied.

II: The initial investment of David is 33.33% of Mano's investment.

$$\text{The required percentage} = (6000/9000) * 100 = 66.66\%$$

This statement is not satisfied.

III: X is an even number.

6000 is an even number

This statement is satisfied.

Therefore, Both I and III are satisfied.

Hence, option A

19. Questions

Answer: D

Let the three-digit number be z

According to the question,

$$z - 399 = 85$$

$$z = 484$$

WKT, the unit digit of a three-digit number is ____ less than its tens place digit

$$8 - 4 = 4$$

First blank = 4

The reversed three-digit number = 484

$$\text{Thus, difference} = 484 - 484 = 0$$

Second blank = 0

Therefore, 4 and 0

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Hence, option D

20. Questions

Answer: E

Let the speed of train A and train B be $7x$ m/s and $11x$ m/s respectively.

According to the question,

$$(d + 300)/7x = (d + 200 + 500)/11x$$

$$11d + 3300 = 7d + 4900$$

$$4d = 1600$$

$$d = 400$$

a). It is a perfect square (true)

b). It is a multiple of 4 (true)

c). It is a three-digit number (true)

Therefore, all (a), (b) and (c)

Hence, option E.

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